Stat: Penalized Zero-Inflated Beta Regression

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Project Summary

Overview

[State the scientific problem in one or two sentences.]

[Explain why the problem matters and what is currently missing.]

[State the central research objective and the overall strategy.]

The central hypothesis is that **[CENTRAL HYPOTHESIS]**.

The project will pursue three complementary aims:

1. **Aim 1:** [FOUNDATIONAL THEORY OR METHOD]
2. **Aim 2:** [STATISTICAL METHODOLOGY]
3. **Aim 3:** [VALIDATION, COMPUTATION, OR SCIENTIFIC APPLICATION]

The expected outcome is [MAIN CONTRIBUTION].

Intellectual Merit

[Explain the fundamental statistical/mathematical knowledge that will be advanced.]

[Describe what is genuinely new relative to existing theory and methods.]

[State the expected theoretical, methodological, or computational contributions.]

Broader Impacts

[Describe concrete impacts through training, software, data, education, interdisciplinary research, dissemination, broadening participation, or other appropriate mechanisms.]

[State how impacts will be assessed.]

Project Description

1. Motivation and Background

Proportion data arise in many scientific settings. Examples 1, 2, 3 etc. Traditionally this type of data has modeled using normal regressions techniques with transformations of the response variable such as the the arcsin square root transformation. However, this method is known to be flawed. Some have even referred to it as "asinine". Other alternatives include using logistic/ binomial regression, however, these require the statistical practitioner to know the denominator (i.e. the n of these proportions). In many settings all we have is the proportion. In these cases, a better

alternative is beta regression (CITATION). Beta regression is well studied (IS IT?) with established packages available in R. Kosmidis and Zeileis [2026]

https://pubmed.ncbi.nlm.nih.gov/21560670/

[Introduce the broad scientific or statistical problem.]

[Explain the current state of knowledge.]

[Identify the specific limitation or gap.]

For example, existing approaches typically rely on [ASSUMPTION], which is problematic when [SETTING]. Existing methods such as [METHODS] provide useful solutions, but do not address [SPECIFIC GAP] [**?**].

The proposed research addresses this gap by developing [NEW FRAMEWORK].

2. Central Research Question

The central research question is:

[CENTRAL RESEARCH QUESTION]

The project is guided by the following hypothesis:

[CENTRAL HYPOTHESIS]

3. Overall Strategy

The proposed work combines [THEORY], [STATISTICAL METHODOLOGY], and [COMPUTATION/APPLICATION].

The aims are deliberately complementary. Aim 1 establishes [FOUNDATION]; Aim 2 develops [METHOD]; and Aim 3 evaluates and demonstrates [CONTRIBUTION].

The project is designed so that failure of any single technical approach will not prevent progress toward the overall objective.

4. Research Plan

4.1 Specific Aims

Aim	Scientific objective	Primary deliverable
1	[OBJECTIVE]	[THEOREM/METHOD]
2	[OBJECTIVE]	[METHOD/SOFTWARE]
3	[OBJECTIVE]	[VALIDATION/APPLICATION]

4.2 Conceptual Framework

The proposed framework is:

The framework links [COMPONENT 1] to [COMPONENT 2] through [MECHANISM].

4.3 Evaluation Strategy

Each aim will be evaluated using a combination of theoretical analysis, simulation, computational benchmarking, and/or scientific validation.

Success will be assessed using prespecified criteria:

- [CRITERION 1]
- [CRITERION 2]
- [CRITERION 3]
- [CRITERION 4]

5. Aim 1: [FOUNDATIONAL THEORY OR METHOD]

Motivation

[Explain the specific gap addressed by Aim 1.]

Research Question

[Specific question.]

Proposed Approach

We will develop [METHOD].

Let Y denote [OUTCOME], X denote [COVARIATES], and θ denote [PARAMETERS]. We consider the general model

$$[MODEL]$$

The principal theoretical objectives are:

1. [OBJECTIVE]
2. [OBJECTIVE]
3. [OBJECTIVE]

Theoretical Development

[Describe identifiability.]

[Describe consistency.]

[Describe asymptotic behavior.]

[Describe efficiency, robustness, or other relevant properties.]

Simulation Study

The simulation study will vary:

Factor	Levels
Sample size	[LEVELS]
Signal strength	[LEVELS]
Missingness/dependence structure	[LEVELS]
Model misspecification	[LEVELS]

Primary performance measures will include [BIAS], [RMSE], [COVERAGE], [POWER], and [COM-

PUTATIONAL COST].

Expected Outcomes

Aim 1 will establish [EXPECTED CONTRIBUTION].

Potential Problems and Alternative Strategies

[Technical risk.]

[Alternative strategy.]

[Criteria for adopting the alternative.]

6. Aim 2: [STATISTICAL METHODOLOGY]

Motivation

[Explain why Aim 1 is insufficient by itself and what methodological problem remains.]

Proposed Method

[Detailed description of estimator/model/algorithm.]

The proposed estimator is

$$\widehat{\theta} = [FORMULA].$$

[Explain estimation.]

[Explain uncertainty quantification.]

[Explain computational implementation.]

Benchmarking

The proposed method will be compared descriptively with established alternatives including [METHOD 1], [METHOD 2], and [METHOD 3].

Evaluation will consider statistical performance, robustness, interpretability, and computational scalability.

Software

The methods will be implemented in an open-source R package, tentatively named `[PACKAGE NAME]`.

The package will include:

- reproducible implementations of all proposed methods;
- simulation utilities;
- diagnostics;
- documentation and examples; and
- vignettes reproducing the principal analyses.

Expected Outcomes

[Expected contribution.]

Potential Problems and Alternative Strategies

[Potential problem and alternative.]

7. Aim 3: [VALIDATION OR SCIENTIFIC APPLICATION]

Scientific Context

[Describe the scientific problem for which the proposed methodology is relevant.]

Data

[Describe data source, sample size, variables, and study design.]

Analysis

[Specify primary analysis.]

[Specify secondary analyses.]

[Specify sensitivity analyses.]

Validation

[Describe external validation, simulation-based calibration, held-out prediction, replication, or other validation strategy.]

Expected Outcomes

[Expected scientific contribution.]

Integration Across Aims

[Explain how Aim 3 tests whether the theory and methodology developed in Aims 1 and 2 produce useful scientific knowledge.]

8. Preliminary Work

Our preliminary work establishes feasibility of the proposed research.

Preliminary Result 1

[Describe preliminary analysis.]

Preliminary Result 2

[Describe second result.]

Method	Bias	RMSE	Coverage
[Existing method]	[ ]	[ ]	[ ]
Proposed method	[ ]	[ ]	[ ]

These preliminary results motivate the proposed development of [METHOD].

9. Intellectual Merit

The intellectual merit of the project rests on [CENTRAL SCIENTIFIC CONTRIBUTION].

Fundamental Contribution

[Explain the fundamental statistical or mathematical question.]

Methodological Innovation

The proposed work advances current methodology by:

1. [INNOVATION]
2. [INNOVATION]
3. [INNOVATION]

Generalizability

[Explain why the results will apply beyond the motivating application.]

The expected contributions include [THEORY], [METHODOLOGY], [COMPUTATION], and SOFTWARE.

10. Broader Impacts

Broader Impacts

Chat GPT write this: The proposed research will develop statistical methodology for analyzing bounded continuous outcomes that exhibit substantial excess zeros. Such outcomes arise in a wide range of scientific and social-scientific settings, including proportions, rates, utilization measures, adherence measures, ecological indices, and other outcomes for which zero represents a meaningful boundary value rather than a missing observation. The proposed development of zero-inflated penalized beta regression will provide researchers with a principled framework for modeling these data while accommodating both the boundary mass at zero and the continuous variation among positive observations.

A primary broader impact of the project will be to improve the accessibility and reliability of statistical methods for investigators who encounter these data in applied research. Existing approaches often require researchers to choose between models that inadequately represent the zero-generating process and models that fail to account for the bounded nature of the nonzero observations. By developing estimation procedures, inferential methods, diagnostics, and computational tools specifically for zero-inflated beta regression with penalization, this project will provide a practical alternative for researchers working with high-dimensional or otherwise challenging datasets. The resulting methods will be designed for implementation in commonly used statistical software, lowering the barrier between methodological development and applied use.

The project will also contribute to statistical education and training. Modern applied statisticians increasingly encounter outcomes that are simultaneously bounded, zero-inflated, and associated with many potential predictors. These characteristics are often treated separately in introductory statistical modeling, despite their frequent occurrence together in contemporary datasets. The proposed work will provide examples and computational materials that can be incorporated into

graduate courses in regression, generalized linear models, high-dimensional statistics, and statistical computing. Reproducible examples will illustrate how model assumptions, regularization, zero inflation, and bounded outcomes interact, providing students with a framework for thinking critically about model selection rather than treating individual modeling techniques as isolated procedures.

An additional impact will be the development and dissemination of open-source computational resources. Software implementing the proposed methodology, together with documentation, examples, simulation code, and reproducible analyses, will be made publicly available. These resources will facilitate independent evaluation of the methods, encourage adoption by applied researchers, and provide a foundation for subsequent methodological extensions. Whenever possible, computational implementations will use established statistical software ecosystems so that the methods can be incorporated into existing analytical workflows.

The project will further support the development of researchers through mentoring and training in statistical methodology, computation, and reproducible research. Students involved in the project will gain experience translating theoretical statistical developments into computational algorithms and practical analytical tools. Training will emphasize not only mathematical and computational skills but also communication of statistical methodology to researchers outside statistics. This combination of methodological depth and applied communication is increasingly important as statistical models are used across scientific disciplines.

Finally, the proposed research has the potential to broaden the role of statistical methodology in scientific investigations where conventional regression models do not adequately reflect the structure of the observed data. Rather than developing a method for a single application, the project addresses a general class of statistical problems. By making the resulting methodology theoretically justified, computationally accessible, and reproducible, the project will contribute to the broader statistical infrastructure available to researchers studying bounded and zero-inflated outcomes.

10.1 Training

[Describe graduate and undergraduate training.]

Students will receive training in:

- statistical theory;
- reproducible research;
- R programming;
- scientific communication; and
- collaborative research.

10.2 Open Science and Software

[Describe planned software and code release.]

All appropriate software will be released under an open-source license, with documentation and reproducible examples.

10.3 Education and Dissemination

[Courses, workshops, public-facing materials, conferences, tutorials, etc.]

10.4 Broadening Participation

[Concrete activities, partnerships, mentoring, or recruitment strategies.]

10.5 Assessment

Activity	Metric	Target
Student training	Students completing research experience	[TARGET]
Software	Downloads/citations/users	[TARGET]
Dissemination	Presentations/workshops	[TARGET]
Publications	Peer-reviewed products	[TARGET]

11. Timeline and Project Management

The project will be conducted over four years.

Activity	Year 1	Year 2	Year 3	Year 4
Aim 1	X	X		
Aim 2		X	X	
Aim 3			X	X
Software	X	X	X	X
Student training	X	X	X	X
Dissemination		X	X	X

Year 1

[Milestones.]

Year 2

[Milestones.]

Year 3

[Milestones.]

Year 4

[Milestones.]

Risk Management

[Describe dependencies and contingency plans.]

12. Results from Prior NSF Support

Award [AWARD NUMBER]

Title: [TITLE]
Amount: $[AMOUNT]
Period: [START–END]

Intellectual Merit

[Results.]

Broader Impacts

[Results.]

Publications and Products

[Publications, software, datasets, presentations, or other products.]

Relationship to Proposed Work

[Explain relationship to the current project.]

Award [SECOND AWARD, IF APPLICABLE]

[Repeat structure.]

If no qualifying prior NSF support exists, replace this section with the appropriate NSF-required statement.

References Cited

Ioannis Kosmidis and Achim Zeileis. Extended-support beta regression for [0, 1] responses. *Journal of the Royal Statistical Society C*, 75(1):139–157, 2026. doi: 10.1093/jrsssc/qlaf039.